

Evaluation of AI-based text-to-speech generators

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Code&Data: https://github.com/avt-imt/text_to_speech

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- ▶ new and various AI text-to-speech (TTS) systems available
 - e.g., Tortoise-TTS v2 [1], VITS [6] or more [2]
- ▶ evaluation of TTS focuses, e.g.
 - on universal perceptual quality dimensions [5]
 - TTS with avatar interaction [12]
 - prediction models [13]
 - **usually:** only individual TTS evaluated or limited comparison
- ▶ **open:** evaluation of different AI-based text-to-speech systems with human ratings and objective metrics

- ▶ 10 different TTS generators (8 female and 2 male voices):
 - open source (Coqui.ai TTS toolbox [2]):
 - ▷ Tacotron2 [15], Glow-TTS [7], SpeedySpeech [19], VITS [6], Fastpitch [8], Overflow [11],
 - ▷ Neural HMM TTS [10], Tortoise-TTS v2 [1] (male)
 - proprietary (Web API):
 - ▷ Google Translate (english), ChatGPT web interface (male)

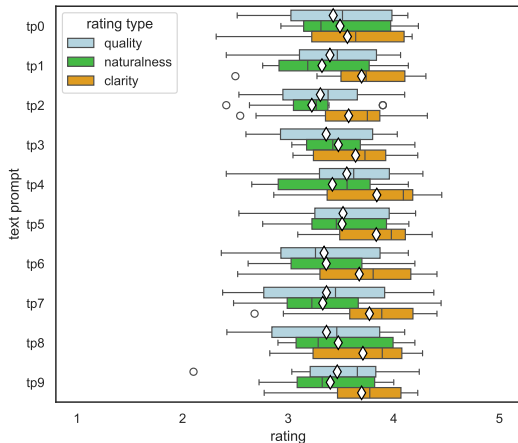
- ▶ 10 text prompts: 3 classes
 - class *D*: tp0 and tp1, from [4], moderate phoneme diversity
 - class *I*: tp2, tp3, tp7, grammatically incorrect but used in everyday life
 - class *E*: tp4, tp5, tp6, tp8, tp9, emotions and vivid mental images

id	class	prompt
tp0	<i>D</i>	Nobody could've imagined that it'd become a breakout hit.
tp1	<i>D</i>	Nowadays, a voice can be cloned with mere minutes of data.
tp2	<i>I</i>	I ain't got no money.
tp3	<i>I</i>	She don't like vegetables.
tp4	<i>E</i>	The sky was ablaze with the colors of the sunset, painting the horizon in hues of orange and pink.
tp5	<i>E</i>	With a heavy heart, he watched her walk away, the weight of unspoken words hanging in the air.
tp6	<i>E</i>	His words were like daggers, each one piercing deeper than the last.
tp7	<i>I</i>	Between you and I, this is a bad idea.
tp8	<i>E</i>	The garden was a riot of colors, with flowers in full bloom swaying gently in the summer breeze.
tp9	<i>E</i>	The thunderstorm raged outside, its fury echoing the turmoil within her heart.

Approach – study design and implementation

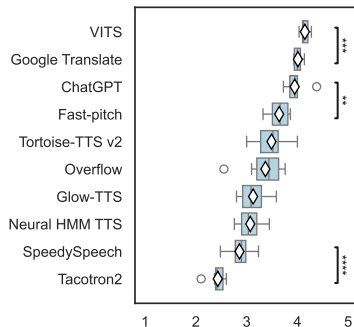
- ▶ overall 100 stimuli (loudness normalized + format unification)
 - 3-4 s each stimuli, FLAC, pyloudnorm [16] (ITU-R BS.1770-4)
- ▶ design of an online study, where participants rated (1-5 ACR-scale):
 - audio quality: technical audio quality,
 - clarity: quality of being easily understandable
 - naturalness: closeness to human voice
- ▶ framework used AVrateVoyager [3]
 - adjustments to include loudness calibration similar to [18]
 - small template and interface changes (welcom screen, instruction)
- ▶ 14 valid users (10 male, 4 female) of 25 from Clickworkers
 - excluded: incomplete ratings, low English skills, hearing aids, outliers (ITU-T P.913)

Results – ratings per text prompt

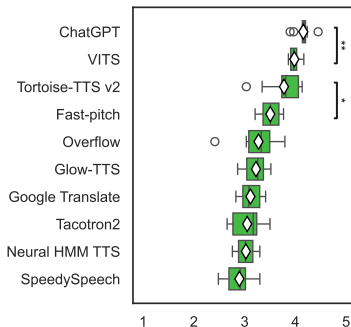


- ▶ mean quality:
 - highest tp4 (3.56), lowest tp2 (3.31).
- ▶ mean naturalness:
 - highest tp5 (3.51), lowest tp2 (3.23)
- ▶ mean clarity:
 - highest tp4 (3.84), lowest tp0 (3.56)
- ▶ overall correlations of dimensions:
 - quality vs. clarity: 0.861 PCC
 - clarity vs. naturalness: 0.712 PCC
 - quality vs. naturalness: 0.688 PCC

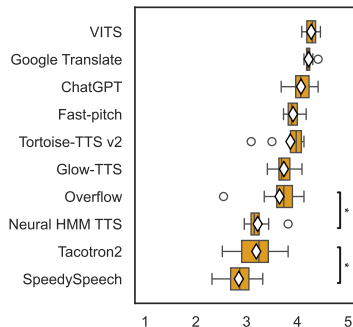
Results – ratings per TTS



(a) quality



(b) naturalness



(c) clarity

- best: VITS, Google Translate, ChatGPT
- worst: Neural HMM, SpeedySpeech and Tacotron2

► features/models:

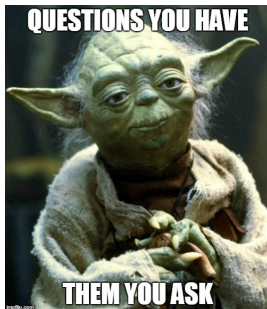
- low-level: frequency bands ($f_0, f_1, \dots$), `speaking_rate`, ...
- high-level:
 - ▷ speech quality (NISQA [13], MOSNet [9], DNSMOS [14], ...),
 - ▷ Meta-CE=Content Enjoyment, Meta-CU=Content Usefulness, Meta-PC=Production Complexity, Meta-PQ=Production Quality [17]

► top-5:

- quality & clarity: $\text{Meta-CU} > \text{Meta-PQ} > \text{Meta-CE} > \text{nisqa_q} > \text{speaking_rate}$
- naturalness: $\text{Meta-CE} > \text{Meta-CU} > \text{Meta-PQ} > \text{speaking_rate} > \text{mean_f0}$

- ▶ core idea and results
 - evaluation of text to speech systems: quality, naturalness, clarity
 - 10 TTS, 10 text prompts = 100 stimuli
 - online study with AVRateVoyager [3]
- ▶ **open source dataset & evaluation**
 - best TTS: VITS, Google Translate, ChatGPT
 - quality and clarity highly connected
 - objective models: Meta Audiobox Aesthetics [17] best for all 3 dimensions
- ▶ future work:
 - comparison with real voices
 - broader range of TTS-systems, voice cloning, more participants

Thank you for your attention



..... are there any questions?

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